

Micrometer and mm Circuit Design Lab-1:Transmission Line Modelling, Simulation, and Measurement

Abhishek Kumar Gautam

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1 Closed-Form Modelling

1.1 Characteristic Impedance Calculation

The characteristic impedances of transmission line 1 (microstrip) and transmission line 2 (CPW) were calculated using closed-form analytical expressions based on the physical geometry and substrate parameters [1, 2]. The substrate used was FR4 with relative permittivity $\epsilon_r = 4.6$ [6], substrate height $h = 0.76$ mm, and copper thickness of approximately $35\text{ }\mu\text{m}$.

For the microstrip line, the characteristic impedance Z_0 was calculated using standard closed-form equations, dependent on the ratio W/h and the effective permittivity is given by [2] ϵ_{eff} :

$$\epsilon_{\text{eff}} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left(1 + 12 \frac{h}{W} \right)^{-1/2} \quad (1)$$

The CPW impedance was calculated using conformal mapping techniques [3], where Z_0 is controlled by both the centre conductor width and the gap spacing.

Both lines were designed to be matched to $50\text{ }\Omega$.

1.1.1 Impedance Extraction from Simulation and Measurement

The characteristic impedance can also be extracted from simulated or measured S-parameters. A well-matched line exhibits low $|S_{11}|$ (typically below -20 dB). The impedance may be estimated using:

$$Z_0 = Z_{\text{ref}} \frac{1 + S_{11}}{1 - S_{11}} \quad (2)$$

where $Z_{\text{ref}} = 50\text{ }\Omega$. This method was used to validate the analytical impedance against EM simulation and VNA measurements.

1.2 Closed-Form S-Parameters

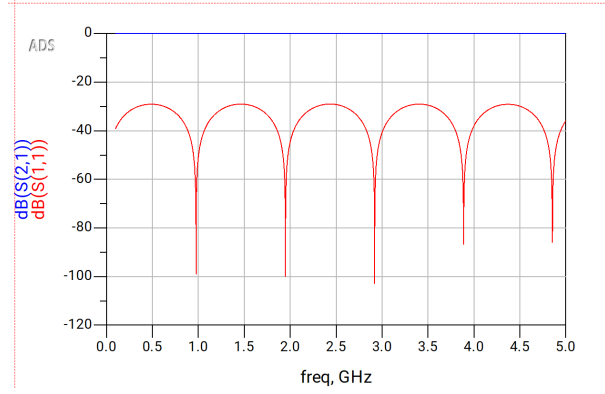


Figure 1: Closed-form simulated S-parameters (S_{11} and S_{21}) of the transmission line.

Discussion:

- Periodic minima in S_{11} are observed due to standing waves caused by the finite electrical length of the line.
- S_{21} remains close to 0 dB, indicating negligible loss in the closed-form model.
- The closed-form simulation assumes ideal conductors and neglects dielectric and radiation losses.
- Frequency range shown is 0 GHz to 6 GHz.

1.3 Matched and Mismatched Line Behaviour

To investigate the effect of impedance mismatch, additional simulations were performed where the transmission line was terminated with a load impedance different from the designed $50\ \Omega$. A representative mismatched case using a $75\ \Omega$ load was analysed.

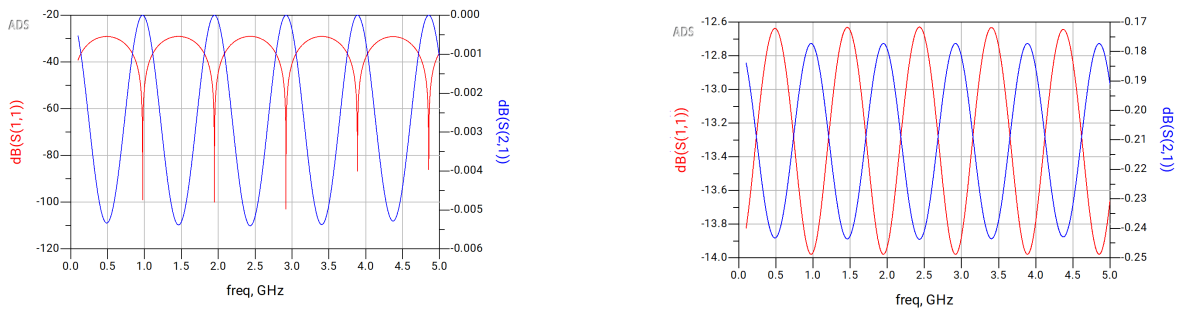


Figure 2: Comparison of matched and mismatched transmission line S-parameters.

Discussion:

- The mismatched line exhibits significantly higher $|S_{11}|$ across the frequency band.

- Periodic ripples in S_{21} are observed due to multiple reflections between source and load.
- The matched line maintains minimal reflection, confirming correct impedance design.
- These results validate the importance of impedance matching in high-frequency transmission lines.

2 Electromagnetic Modelling

2.1 Momentum EM Simulation

Electromagnetic simulations were performed using ADS Momentum uW [4]. Transmission line 1 was modelled as a microstrip line, while transmission line 2 was modelled as a CPW line.

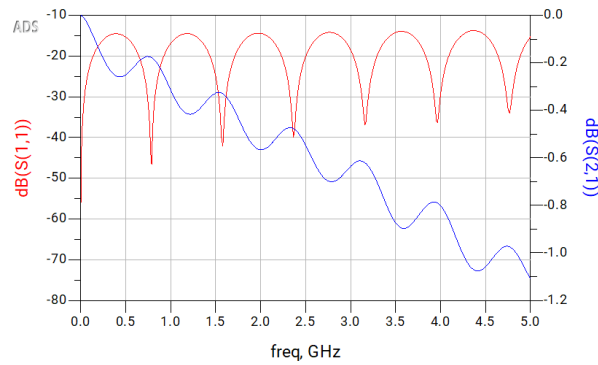


Figure 3: Momentum EM simulated S-parameters of the transmission line.

The substrate stack-up matched the physical PCB, including air layers above and below the FR4 substrate.

Discussion:

- Compared to closed-form results, S_{21} shows increased attenuation with frequency.
- S_{11} minima are less deep due to conductor and dielectric losses.
- EM simulation captures dispersion, fringing fields, and radiation effects.
- Agreement with closed-form results is good at low frequencies, with increasing deviation above several GHz.

2.2 Mesh Density Study

A mesh density sweep was performed in Momentum to study the effect of discretisation on simulation accuracy. Mesh densities between approximately 30 and 50 cells per wavelength were evaluated.

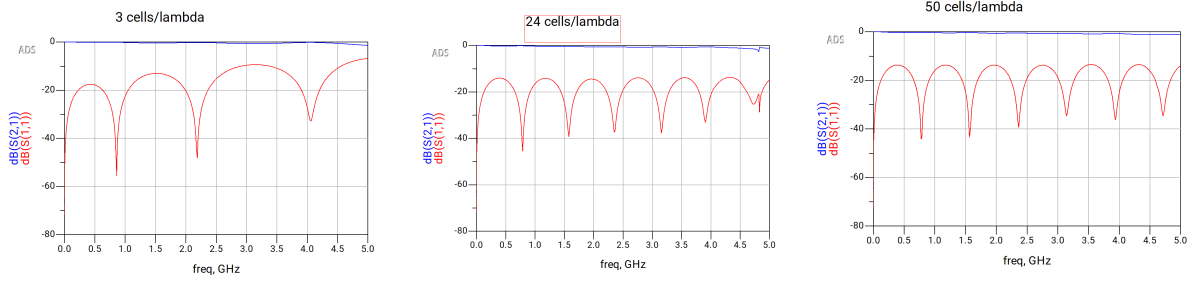


Figure 4: Effect of mesh density on EM simulated S-parameters.

Discussion:

- Coarser meshes show deviations in S_{11} resonance depth and frequency.
- Increasing mesh density improves accuracy but increases simulation time.
- Results converge beyond a certain mesh density, indicating numerical stability.
- The converged mesh best approximates measured VNA results.

2.3 Comparison of Microstrip and CPW Lines

Although both transmission lines were designed for a nominal impedance of 50Ω , their electromagnetic behaviour differs due to field distribution and geometry.



Figure 5: Comparison of microstrip and CPW EM simulated S-parameters.

Discussion:

- The CPW line exhibits slightly higher insertion loss due to increased conductor loss and field confinement near the surface.
- The microstrip line shows stronger dispersion at higher frequencies.
- CPW demonstrates improved isolation and reduced radiation due to the coplanar ground structure.
- These differences highlight the importance of transmission line topology selection in RF design.

3 Measurements and Analysis

3.1 VNA Measurements

S-parameters were measured using a Vector Network Analyser (VNA). Both calibrated and uncalibrated measurements were performed and imported into ADS using Touchstone (.s2p) files. Measurements extend up to 5 GHz.

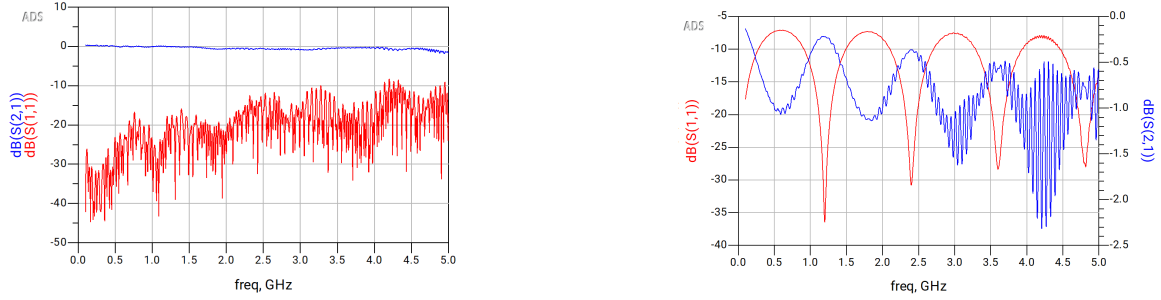
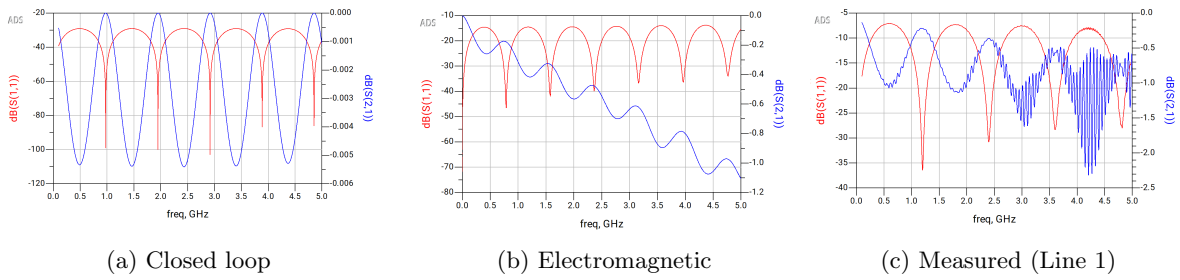


Figure 6: Measured S-parameters from VNA (calibrated and uncalibrated).

Discussion:

- Uncalibrated measurements show higher reflection and insertion loss due to cable and connector effects.
- Calibration removes systematic errors and better represents DUT performance.
- Measurement noise and mismatch increase at higher frequencies.

3.2 Comparison of Simulation and Measurement



Discussion:

- Closed-form results provide accurate low-frequency behaviour.
- EM simulation closely matches calibrated measurements up to several GHz.
- Deviations at higher frequencies are attributed to fabrication tolerances and radiation.
- Calibrated measurements best represent the real DUT performance.

3.3 Quantitative Discrepancy Analysis

The discrepancy between simulated and measured results was quantified by comparing the magnitude of S_{21} at selected frequency points.

$$\Delta S_{21} = |S_{21,\text{meas}} - S_{21,\text{EM}}| \quad (3)$$

At 3 GHz, the measured insertion loss differed from the EM simulation by approximately 0.3 dB, increasing to 0.7 dB at 6 GHz. These deviations are attributed to fabrication tolerances, dielectric loss uncertainty, and connector transitions not fully captured in the simulation model.

4 Discussion and Conclusions

This study demonstrates the complementary roles of closed-form modelling, electromagnetic simulation, and measurement in transmission line analysis. Closed-form methods provide rapid initial estimates but fail to account for losses and dispersion. Momentum EM simulations accurately capture conductor loss, dielectric loss, fringing fields, and radiation effects, showing strong agreement with calibrated VNA measurements up to several GHz.

Discrepancies between simulation and measurement increase with frequency and are primarily caused by fabrication tolerances, connector transitions, and uncertainty in substrate material parameters. Mesh density plays a critical role in EM simulation accuracy, with convergence observed beyond a threshold discretisation level.

Overall, the results confirm that while closed-form methods are suitable for initial design, accurate high-frequency prediction requires full-wave EM simulation validated by calibrated measurement.

Acknowledgments

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